

Preethivardhan Jayabal - Sr. Design Engineer (NPD)

M.Tech, Product Design and Manufacturing | Lean Six Sigma Black Belt

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Summary

Senior Mechanical Design Engineer with 4+ years of experience in New Product Development, specializing in CAD modelling, engineering drawings, GD&T, and Design for Manufacturability (DFM). Currently leading a small NPD team and managing projects from concept to deployment. Lean Six Sigma Green Belt certified with proven experience in process improvement, digital production tracking, and QR-based traceability systems.

Experience

Sr. Design Engineer – New Product Development

Metal Scope (India) Pvt. Ltd., Puducherry | Jan 2022 – Present

- Lead the design and development of mechanical systems, modular assemblies, and structural components from concept to deployment of modular building systems.
- Develop 3D CAD models, manufacturing drawings, BOMs, and engineering documentation using SolidWorks.
- Apply GD&T (ASME Y14.5), tolerance stack-up analysis, and DFM principles to ensure manufacturability and assembly feasibility.
- Execute design changes through ECR/ECO processes while collaborating with manufacturing, procurement, and project stakeholders.
- Utilize FDM 3D printing for rapid prototyping and design validation.
- Lead a small New Product Development team, driving design reviews, engineering coordination, and project execution.

Process Improvement & Digital Transformation (Additional Responsibility)

- Optimized manufacturing workflows and industrial layouts, contributing to a 25% productivity improvement.
- Led a Lean Six Sigma Green Belt project focused on production process digitalization and workflow optimization.
- Developed and implemented QR-based finished goods traceability and digital production tracking systems.
- Presented process improvement initiatives at the National Convention on Quality Control (NCQC).

Education

M.Tech – Product Design and Manufacturing

Pondicherry Engineering College (2019 – 2021)

- Conducted FEA analysis of elastomeric O-ring, focusing on sealing performance and hyperelastic material modeling.

B.Tech – Mechanical Engineering

Rajiv Gandhi College of Engineering and Technology (2015 – 2019)

Technical Skills

CAD Tools:

SolidWorks | AutoCAD | CATIA (Learning)

Design & Analysis:

ANSYS | Basic FEA | GD&T (ASME Y14.5) | Tolerance Stack-up | Engineering Drawings | BOM Preparation

Manufacturing:

Design for Manufacturability (DFM) | Fabrication Coordination | Cost Optimization

Tools:

ERP Systems | MS office suite

Methods:

Lean Six Sigma | DMAIC | Process Optimization | Workflow Improvement | Digital Transformation

Certifications

Lean Six Sigma Green Belt – MSME

Lean Six Sigma Green Belt – MSME

SolidWorks Associate (CSWA)

SolidWorks Advanced Professional – Weldments

Additive Manufacturing Associate

DraftSight – 2D Design & Drafting

ANSYS Workbench – CAD Solutions

Key Achievements

- Led implementation of QR-based production traceability system.
- Presented the Lean Six Sigma project at the National Convention on Quality Control (NCQC).
- Achieved 25% productivity improvement through workflow optimization.
- Currently leading a New Product Development team.

Key Strengths

Lean Six Sigma (Black Belt) | DMAIC Methodology | Process Optimization | Digital Process Implementation | Mechanical Design & CAD Modelling | GD&T & Tolerance Analysis | Cross-functional Coordination | Engineering Documentation | Problem Solving & Continuous Improvement | Team Coordination & Task Management